

# Classification of the 2006 formula language

On S. A. Alexander, *Formulas for Computable and Noncomputable Functions* (2006)

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**Status.** A proposed answer with a complete hand argument. Definition 2 and the question were checked visually against the published PDF, whose text encoding is corrupt. There has been no independent review and no proof-assistant check. Targeted searches did not locate an earlier explicit answer to this formulation. The argument uses standard arithmetical definability, so no originality claim is made for its techniques.

**Source.** S. A. Alexander, *Formulas for computable and noncomputable functions*, Rose-Hulman Undergraduate Mathematics Journal 7(2), 2006. [Record](#). The question addressed is the direction posed immediately after Corollary 9 (printed p. 23): which functions does the formula language of Definition 2 describe?

**Answer.** Exactly the partial functions with **arithmetical graphs**.

## Exact statement and conventions

Let  $F$  be the inductively generated class in **Definition 2**, printed p. 5. Its functions have finite positive arity and natural-number outputs, and may be partial. It starts with 0, 1, projections, Kronecker delta, addition, multiplication, absolute difference, and natural exponentiation with  $0^0 = 1$ . It closes under composition, finite summation with a formula-defined natural bound, and infinite summation.

An infinite sum is defined exactly when every summand is defined and the series converges to a finite natural number. Composition has its usual strict partial-function semantics. Formulas are finite expressions generated by these operations; an infinite summation binds a variable in one such expression, not an arbitrary independently specified infinite list of formulas.

**Theorem.** For each  $k \geq 1$ , a partial function  $f : \mathbb{N}^k \rightarrow \mathbb{N}$  belongs to  $F$  if and only if its graph

$$\Gamma_f = \{(\mathbf{x}, y) : f(\mathbf{x}) \text{ is defined and equals } y\}$$

is arithmetically definable. In particular, the total members are exactly the total functions with arithmetical graphs. “Arithmetical” here permits a finite formula of first-order arithmetic and ordinary natural-number parameters, not a set oracle or arbitrary real parameter.

## 1. Every formula has an arithmetical graph

Induct on its construction. Every atomic function has an arithmetical graph. Composition is expressed by existentially quantifying intermediate values and requiring the component graph relations; this also handles undefined inputs correctly.

For finite sums, quantify the natural bound  $b$ , require the bound function's graph relation, and quantify a code for a finite list of length  $b + 1$ . Require its  $i$ -th entry to be the value of the summand at  $i$  for every  $i \leq b$ , and its sum to be  $y$ . Finite lists, their entries, lengths and sums have standard arithmetical encodings. This is therefore one arithmetical formula, including definedness of every required term.

For an infinite sum  $h(\mathbf{x}) = \sum_{i=0}^{\infty} u(\mathbf{x}, i)$ , nonnegative integer values are crucial: convergence to a finite value is equivalent to all terms being defined and only finitely many being nonzero. Thus its graph is expressed by

$$\begin{aligned} \exists N \exists s [ & \text{len}(s) = N \wedge \text{sum}(s) = y \\ & \wedge \forall i < N \Gamma_u(\mathbf{x}, i, (s)_i) \\ & \wedge \forall i \geq N \Gamma_u(\mathbf{x}, i, 0) ]. \end{aligned}$$

Here  $s$  codes a finite list; the sum of an empty list is zero. The notation abbreviates arithmetically definable relations, so only first-order number quantifiers are involved. If a summand is undefined at any index, no such witness exists. The displayed condition is both necessary and sufficient, completing the induction.

The finiteness of each formula matters. Different formulas can reach arbitrarily high finite levels of the arithmetical hierarchy, but no one finite construction invokes an unbounded tower of new quantifier levels.

## 2. Indicators of arithmetical predicates belong to the language

This is the content of the paper's Theorem 8/Corollary 9, but a direct argument makes our dependence clear.

Atomic equality of arithmetic terms is represented by Kronecker delta; terms and Boolean combinations are expressible using the atomic operations. If  $b(\mathbf{x}, i) \in F$  is a total  $\{0, 1\}$ -valued indicator, define

$$E_b(\mathbf{x}) = \sum_{i=0}^{\infty} b(\mathbf{x}, i) \delta \left( 1, \sum_{j=0}^i b(\mathbf{x}, j) \right).$$

Exactly the **first** true instance contributes 1; all other summands are zero. When there is no true instance, every summand is zero. This is a total formula for  $\exists i b(\mathbf{x}, i) = 1$ . Complement is  $|1 - b|$ , and universal quantification follows by complementing an existential quantifier. Induction over any finite arithmetic formula produces its total indicator. Multiple variables cause no difficulty; alternatively, use an arithmetically definable tuple coding.

## 3. Recover a partial function from its graph indicator

Let  $\Gamma_f$  be arithmetical, and let  $b(\mathbf{x}, j)$  be its total indicator, supplied by the previous step. Put

$$H(\mathbf{x}) = \sum_{n=0}^{\infty} \delta \left( 0, \sum_{j=0}^n b(\mathbf{x}, j) \right).$$

If  $f(\mathbf{x}) = m$ , the inner sum is zero exactly for  $n < m$ . Consequently the outer series contains exactly  $m$  ones, followed by zeros, and equals  $m$ . This includes  $m = 0$ , where every summand is zero.

If  $f(\mathbf{x})$  is undefined, the inner sum is always zero and the outer series is  $1 + 1 + \dots$ , so  $H$  is undefined. Thus  $H = f$  with exactly the same domain. It uses only the allowed finite and infinite sums and delta, completing the reverse inclusion.

## Consequences and limits

The paper proves that the language can describe every computable function and indicators of noncomputable arithmetical sets. The classification identifies the upper boundary too. No claim about practical evaluation follows: a formula denoting an integer does not come with an algorithm for recognizing that all future summands vanish. Even a series with at most one nonzero term can encode an undecidable existential assertion.

The proof classifies **Definition 2**, not the enlarged real-valued language of §5. Arbitrary real-valued summands no longer satisfy the finite-support characterization used above. It also does not classify formula length, summation depth or computational cost under a promised effective tail bound. Those are natural subsequent questions, requiring a precise resource model and a separate literature review.